Tutorial 3 - Practical, real-world coding examples for Functions, Modules and Object-Oriented Programming.

Section A: Multiple Choice Questions (MCQ)

1. What is the main benefit of using functions?

A. Reduce memory usage
B. Reuse code and improve readability
C. Increase execution speed
D. Avoid loops

Answer: B

2. Which keyword defines a function?

A. func
B. def
C. function
D. define

Answer: B

3. What does a function return if no return is specified?

A. 0
B. None
C. Error
D. Empty string

Answer: B

4. How do you import a module named utils?

A. include utils
B. using utils
C. import utils
D. load utils

Answer: C

5. How do you call function calculate() from module finance?

A. finance->calculate()
B. finance.calculate()
C. calculate.finance()
D. finance:calculate()

Answer: B

6. Why are modules useful?

A. Reduce typing
B. Organize code into reusable files
C. Improve CPU speed
D. Replace functions

Answer: B

7. Which is correct function call?

A. add 5,3
B. add(5,3)
C. add[5,3]
D. add{5,3}

Answer: B

8. What is parameter?

A. Output of function
B. Input to function
C. Loop variable
D. Module

Answer: C

9. Which is built-in module?

A. myfile
B. math
C. student
D. custom

Answer: B

10. What is good practice for modules?

A. Put everything in one file
B. Use many print statements
C. Separate logic into files
D. Avoid functions

Answer: C

Section B: Programming Questions (Real-World Examples)

1. Student Grade Calculator

```
def calculate_grade(score):
    if score >= 80:
        return "A"
    elif score >= 60:
        return "B"
    else:
        return "C"

def main():
    student_name = "Alice"
    score = 75
    grade = calculate_grade(score)
    print(student_name, "Grade:", grade)

if __name__ == "__main__":
    main()
```

Question: What is the output?

Answer: Alice, Grade:, B

2. Simple E-commerce Price Calculation

```
def calculate_total(price, quantity):
    return price * quantity

def main():
    item = "USB Cable"
    price = 5
    quantity = 3
    total = calculate_total(price, quantity)
    print("Total price for", item, "is $", total)

if __name__ == "__main__":
    main()
```

Question: What is printed?

Answer: Total price for USB Cable is $ 15

3. Module Example (Utility Functions)

```
# utils.py
def celsius_to_fahrenheit(c):
    return (c * 9/5) + 32

# main.py
import utils

def main():
    temp_c = 25
    temp_f = utils.celsius_to_fahrenheit(temp_c)
    print("Temperature in F:", temp_f)

if __name__ == "__main__":
    main()
```

Question: What is the output?

Answer: Temperature in F: 77

4. Login System Simulation

```
def check_login(username, password):
    if username == "admin" and password == "1234":
        return True
    return False

def main():
    user = "admin"
    pwd = "1234"

    if check_login(user, pwd):
        print("Login successful")
    else:
        print("Login failed")

if __name__ == "__main__":
    main()
```

Question: What is printed?

Answer: Login successful

5. Simple Data Processing (Average Score)

```
def calculate_average(scores):
    total = sum(scores)
    return total / len(scores)

def main():
    scores = [70, 80, 90]
    avg = calculate_average(scores)
    print("Average score:", avg)

if __name__ == "__main__":
    main()
```

Question: What is the output?

Answer: Average score: 76.6667